

SEGA JET ROCKET (1970)

Electromechanical Restoration Guide

Historic flight / shooter — practical notes for public & educational display

Prepared for **Coin Op 4 Charity** / Arcade Angels

Preserving an important piece of arcade history for youth and community spaces

1. What This Machine Is

Sega Jet Rocket (1970) is an electromechanical (EM) arcade game — no computer, no video screen. The player “flies” a jet through a scrolling landscape viewed via mirrors and shoots illuminated targets. It is widely regarded as an early open-world and first-person shooter design. The cabinet is large and heavy; a person can actually step inside parts of it. Because so few complete working examples remain, careful restoration has high educational and historical value.

2. Safety & Working Principles

- Unplug the machine before any internal work.
- This is 1970 electromechanical technology: motors, relays, solenoids, and higher-voltage circuits. Treat every wire as live until proven otherwise.
- Capacitors in the power and amplifier sections can hold charge. Discharge carefully.
- The scrolling canvas, mirrors, and black-light system create the visual effect — keep optics clean and aligned.
- Document every step with photos. These machines are rare; reversible work and clear notes help future volunteers.

3. Cabinet Exterior & Access

Clean the exterior thoroughly. Inspect for structural wood or metal damage — the cabinet is large and heavy, so integrity matters for safe transport and public display. Clean the viewing glass and marquee plastic. Replace or service the fluorescent marquee lamp and starter so the machine looks alive even when idle. Confirm all access panels and doors open freely and can be secured.

4. Optics, Canvas & Target System

The heart of the visual experience is a continuous scrolling canvas (landscape) viewed through mirrors, with targets illuminated by lights beneath. Foam rubber on the rollers commonly degrades into crumbly residue after decades — clean it completely and replace the foam if needed so the canvas tracks smoothly. Clean the canvas itself carefully; avoid harsh solvents that could damage printed scenery. Inspect and clean the black lights and any associated fixtures. The target-detection system uses conductive pads on the underside of the canvas contacting finger-like prongs; clean these contacts and confirm continuity. Score is often indicated by a bank of small lamps driven by a stepper unit — corroded sockets are extremely common. Replace bad sockets and bulbs so scoring is reliable and visible.

5. Timer Motor & Gearbox

The timer mechanism uses a motor and multi-stage gearbox. Gears frequently become packed with old grease and dirt, causing binding or stalling. Disassemble stage by stage (photograph the order), clean every gear and shaft thoroughly, re-lubricate with appropriate light grease, and reassemble. Check for missing or loose bearings. Free, smooth rotation of the timer is essential for consistent game length.

6. Control Board, Relays & Switches

The control board contains a modest number of relays and switches (far fewer than a typical pinball machine of the era). Remove the board if possible for thorough cleaning. Vacuum and brush away dust. Clean every relay contact and switch with appropriate contact cleaner and burnishing tools. Manually actuate relays and verify clean make/break. Re-seat the board carefully — wood swelling over 50+ years can make the slide-in fit very tight.

7. Sound System (Amplifier & Mixer Boards)

Original sound is often weak, distorted, or full of hum because the electrolytic capacitors are 50+ years old and dried out. Recap the amplifier board and any mixer/tone boards. Also inspect and replace the larger power-supply filter capacitors that feed the audio section. After recapping, adjust the pots that balance the various sound effects according to any labels or notes still present on the board. Clean speaker connections and confirm the speaker cone is intact.

8. Player Controls & Mechanical Linkages

The steering / stick mechanism is largely mechanical and linked to the mirror positioning system. Clean, lubricate, and free any sticky linkages. There is typically a solenoid that locks or releases the control when the game is not active — inspect the solenoid, linkage, and return spring. Confirm the fire button and any other player switches clean and register reliably. Adjust mechanical stops and springs so the mirror movement feels smooth and proportional to the player's input.

9. Power & General Electrical

Inspect the main transformer and power wiring for heat damage or brittle insulation. Replace any questionable capacitors in the power section. Verify fuses are the correct rating and that holders are clean. Trace and repair any broken or highly oxidized wiring, especially around high-current solenoids and motors. A clean, well-supported harness reduces intermittent faults that are hard to diagnose later.

10. Final Testing & Public Display Notes

- Full game cycle runs from coin/start through timer expiration without stalling
- Canvas scrolls smoothly and targets illuminate correctly when hit
- Score lamps advance and display properly
- Sound effects are clear and free of excessive hum
- Controls feel smooth; mirror tracks player input without binding
- Marquee and interior lighting present a clean, inviting appearance
- Cabinet is structurally sound, clean, and safe for public approach
- All access panels secured; no loose parts that could fall during play

For youth or community display, add a short interpretive card explaining that this is a 1970 electromechanical design — no computer, all motors, lights, relays, and clever optics. That context turns a rare machine into a teaching tool about how arcade games worked before microprocessors.

This guide is intended for educational and nonprofit restoration use by Coin Op 4 Charity and partner organizations. Jet Rocket is a scarce and historically significant machine; careful, documented work helps keep it alive for the next generation.

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